



Solution Guide: Interpret Protocols

Completing this activity required the following steps:

- Opening the log file to view the various log records
- Using a web tool to convert the binary data to readable text
- Analyzing the data from the text to determine the protocol being used

Open the log file and note that it contains multiple log records.

Convert the binary data for each log record to a readable format with a web tool of your choosing.

- There are many web tools available for converting binary data to readable text. An easy-to-use tool is the [String Functions binary to string converter](#).
- Go to the site, copy the binary log records, one at a time, into the top field and click "Convert!"

Log Record 1

Log Record 1 after converting:

Unset

```
GET / HTTP/1.1
Host: widgets.com
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.117
Safari/537.36
Accept:
text/html,application/xhtml+xml,application/xml;q=0.9,image
/webp,image/apng,*/*;q=0.8
```

```
Accept-Encoding: gzip, deflate
Accept-Language: en-US,en;q=0.9,nb;q=0.8
```

- The protocol is HTTP.

Log Record 2

Log Record 2 after converting:

Unset

```
File Transfer Protocol (FTP)
230 Login successful.\r\n
Response code: User logged in, proceed (230)
Response arg: Login successful.
```

- The protocol is FTP.

Log Record 3

Log Record 3 after converting:

Unset

```
TLSv1.2 Record Layer: Application Data Protocol:
http-over-tls
Content Type: Application Data (23)
Version: TLS 1.2 (0x0303)
Length: 56
Encrypted Application Data:
d03ff41452da9e9c3ec76cbeb35e8ffc1f64bf80f512924a?
```

- The protocol of TLS or HTTP are both acceptable answers, as https-over-tls is https.

Log Record 4

Log Record 4 after converting:

Unset

Domain Name System (query)

Transaction ID: 0x18b6

Flags: 0x0100 Standard query

0... .. = Response: Message is a query

.000 0... .. = Opcode: Standard query (0)

.... ..0. = Truncated: Message is not

truncated

.... ..1 = Recursion desired: Do query

recursively

.... ..0.. = Z: reserved (0)

.... ..0 = Non-authenticated data:

Unacceptable

Questions: 1

Answer RRs: 0

Authority RRs: 0

Additional RRs: 0

Queries

applegate.com: type A, class IN

[Response In: 623]

- The protocol is DNS.

Log Record 5

Log Record 5 after converting:

Unset

Address Resolution Protocol (request)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

```
Sender MAC address: Technico_65:1a:36
(88:f7:c7:65:1a:36)
Sender IP address: 10.0.0.1
Target MAC address: 00:00:00_00:00:00
(00:00:00:00:00:00)
Target IP address: 10.0.0.6`
```

- The protocol is ARP.

Bonus Log Record

Bonus Log Record after converting:

Unset

HCI H4

[Direction: Unspecified (0xffffffff)]

HCI Packet Type: HCI Command (0x01)

HCI Command - Read Local Supported Features

Command Opcode: Read Local Supported Features (0x1003)

Parameter Total Length: 0

[Response in frame: 4]

[Command-Response Delta: 4.181ms]

- HCI stands for host controller interface, which is used via Bluetooth.